Open Wearables Migration Plan Comparison

Old Migration Plan vs New Execution Checklist

Executive Summary

After comparing both documents, the new checklist is significantly more than an updated version of the original migration plan.

The architecture remains largely the same, but the new checklist introduces multiple entirely new engineering disciplines that were either absent or only briefly mentioned in the original document.

The old document answers:

> What should we build?

The new document answers:

> Exactly how do we build it, verify it, test it, deploy it, monitor it, roll it back, and prevent data corruption?

Overall Coverage Comparison

Area	Old Plan Coverage	New Checklist Coverage
Architecture Design	95%	100%
Backend Design	90%	100%
Frontend Design	80%	100%
Rollout Strategy	75%	100%
Testing Strategy	25%	100%
Data Integrity Controls	15%	100%
Failure Mode Analysis	10%	100%
Operational Safety	10%	100%
Production Readiness	20%	100%
Deployment Governance	5%	100%
CI / Review Process	0%	100%

Area	Old Plan Coverage	New Checklist Coverage
Implementation Tracking	0%	100%

Estimated overall completeness:

- Old Plan: ~55-60%
- New Checklist: ~95-100%

Major Additions in the New Checklist

1. Data Corruption Prevention Layer (Entirely New)

The old document acknowledged data ownership and Mongo/Postgres separation but did not deeply address corruption risks.

The new checklist introduces:

- WearablesSafetyConfig
- Local Mongo enforcement
- Boot-time Mongo validation
- DateBucketKey
- Date format enforcement
- Timezone handling
- DST handling
- Date bucket validation
- Regression tests for date handling

These protections were added because previous testing actually corrupted Mongo data.

This entire safety layer is essentially new.

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2. Formal Data Integrity Framework (Entirely New)

The checklist introduces a dedicated risk matrix:

- Identity mapping failures
- Duplicate delivery failures
- Unit conversion failures
- Timezone attribution failures
- Partial write failures
- Multi-source overwrite failures
- Event ordering failures

- Schema drift failures

Each risk now has:

- mitigation
- tests
- monitoring requirements

The old document did not provide this structure.

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3. Comprehensive Webhook Reliability Model

The old document described:

- webhook receiver
- signature verification
- idempotency

The new checklist expands this into:

- duplicate handling
- stale timestamp handling
- malformed signature handling
- replay protection
- webhook audit log requirements
- TTL index validation
- reconciliation recovery
- webhook metrics

The operational depth is substantially higher.

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4. Reconciliation Architecture

The old document mentioned reconciliation.

The new checklist defines:

- OwReconciliationJob
- repair logic
- repair tests
- drift detection
- metrics
- protection against repairing into wrong environments

This is a major expansion.

5. Full Test Architecture

This is probably the single largest addition.

The old document simply stated that testing would exist.

The new checklist specifies:

Backend:

- mapper tests
- webhook tests
- repository tests
- controller tests
- reconciliation tests
- parity tests

Frontend:

- provider tests
- coordinator tests
- API tests
- routing tests
- silent push tests
- initialization tests
- read-path tests

Operational:

- parity reports
- rollout validation

This is an entirely new layer of engineering rigor.

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6. Explicit Failure Mode Catalog

The checklist contains 23 identified failure modes.

Examples:

- wrong Mongo instance
- date corruption
- duplicate webhook delivery
- HTTP/2 request body loss
- token field drift

- multi-source overwrite
- timezone drift
- SDK ABI breaks
- silent push failures

The old document did not have anything comparable.

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7. Production Governance Layer

Entirely new.

Includes:

- PR limits
- reviewer requirements
- CI requirements
- rollout gates
- rollback requirements
- feature-flag governance
- branch strategy

This is effectively a release-management framework.

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8. Actual Implementation Status Tracking

The old plan assumed future implementation.

The new checklist tracks:

- what is already merged
- what is complete
- what remains
- what tests have passed
- what is still blocked

This transforms the document from a proposal into a project execution dashboard.

Phase Coverage Analysis

Phase 0 – Preparation

Old Plan Coverage: ~30%

The old document covered infrastructure setup but not:

- local Mongo
- baseline snapshots
- smoke testing
- environment verification

Most of this phase is new.

Phase 1 – Safety Rails

Old Plan Coverage: ~10%

Almost entirely new.

Includes:

- boot guard
- date bucket enforcement
- startup protection
- corruption prevention

Phase 2 – Backend Pipeline

Old Plan Coverage: ~85%

Most architecture already existed.

The checklist adds:

- test coverage
- verification
- metrics
- hardening
- implementation status

Phase 3 – Mobile Integration

Old Plan Coverage: ~75%

The architecture already existed.

The checklist adds:

- test suites
- routing validation
- silent push validation
- read-path protection

Phase 4 – Parity Validation

Old Plan Coverage: ~20%

This is mostly new.

Includes:

- parity endpoint
- parity reports
- acceptance thresholds
- drift detection

Phase 5 – Internal Rollout

Old Plan Coverage: ~70%

The concept existed.

The checklist adds:

- monitoring requirements
- rollback rehearsal
- launch criteria

Phase 6 – Expansion & Cleanup

Old Plan Coverage: ~90%

Very similar.

Only minor additions in the checklist.

Biggest Architectural Difference

Interestingly:

The architecture itself barely changed.

The largest difference is that the new checklist assumes:

"Architecture is not the hard part."

The hard part is:

- correctness
- testing
- rollout safety
- corruption prevention
- recovery
- observability

The new checklist spends more effort preventing failure than designing the architecture itself.

That is why it is dramatically larger despite describing essentially the same target system.

Final Assessment

The old migration plan provides approximately:

- 90-95% of the target architecture

but only about:

- 50-60% of the actual implementation and operational plan.

The new checklist provides:

- 95-100% of the implementation plan
- 95-100% of the testing plan
- 95-100% of the rollout plan
- 95-100% of the safety and recovery plan

In practice, the new checklist should be viewed as:

Old Architecture Plan

+ Data Integrity Program

+ Test Strategy

+ Release Strategy

+ Operational Runbook

+ Production Hardening Guide

+ Implementation Status Tracker

rather than merely a revised version of the old document.